Neha Navarkar

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Summary

Robotics engineer with 2+ years of hands-on experience in navigation, perception and control, from design through hardware deployment. Built a visual-inertial navigation stack for a mobile manipulator and now works on safe control of an underactuated balancing robot. Co-author on two robotics preprints. Authorized to work in the US (OPT, eligible for 24-month STEM extension).

Education

New York University — New York, NY
M.S. in Robotics and Mechatronics Engineering — GPA: 3.7/4.0 — *Aug 2024 – May 2026*

D Y Patil College of Engineering, Akurdi — Pune, India
B.E. in Electronics Engineering — GPA: 9.18/10 — *2019 – 2023*

Skills

Autonomy & Perception: ROS2, ZED 2i VIO, voxel mapping, A*, pure pursuit, AprilTag pose estimation, OpenCV, point clouds, Rerun, Viser
Control: Nonlinear MPC, system identification, reaction-wheel and balance control, PID, FOC motor control
Robot Learning: Isaac Lab, PPO/RL, sim-to-real transfer, PyTorch
Embedded & Hardware: ESP32-S3, CAN, ODrive, Jetson, Raspberry Pi, UART/I2C/SPI, Fusion 360
Languages & Tools: Python, C++, C, Docker, Linux, Git, Gazebo

Experience

Computer Vision Contractor — Pipeline Development — New York, NY
Confidential client — *Oct 2026 – Present*
- Building an end-to-end multi-person detection and tracking pipeline with client-defined labels per tracked person.
- Mapping people to spatial coordinates with zones/POIs; delivering structured outputs, documented limits and handoff code.

Research Assistant — ASAS Labs (Agile Safe Autonomous Systems) — New York, NY
NYU Tandon & CCNY — *2026 – Present*
- Second author, arXiv:2610.16223: event-driven safe control of a one-wheel balancing robot with a lead-screw weight shift and a reaction wheel for yaw, whose torque pulses are gated by static friction.
- Wrote ESP32-S3 C++ firmware driving an ODrive FOC controller over CAN; fixed jitter from blocking serial I/O.
- Identified actuator dynamics via step-response least squares; built a nonlinear MPC with slew and travel limits.
- Built overhead-camera AprilTag localization (Orbbec Femto Bolt); diagnosed mirrored heading and 180° ambiguity.

Research Assistant — CILVR Lab (Generalizable Robotics & AI) — New York, NY
New York University — *May 2025 – May 2026*
- Built the navigation stack for YOR on a ZED 2i and Jetson/NUC: visual-inertial odometry, 5 Hz voxel mapping, weighted A* planning, pure-pursuit control.
- Designed a two-layer log-odds voxel map (working and persistent memory) that decays unconfirmed voxels and weights misses over hits; inspected drift in Rerun and Viser.
- Reached 12 mm odometry accuracy over 10 loops, <1 s replanning, and 16 mm end-effector deviation during base motion.

Graduate Assistant — VIP STEP & MakerSpace — New York, NY
New York University — *Jan 2025 – May 2026*
- Designed bipedal gait patterns for uneven terrain; wrote SOPs and trained 50+ students on the UR5 arm.

Graduate Apprentice — Chassis Systems — Chakan, India
Bosch — *Sep 2023 – Aug 2024*
- Won 1st place at Bosch's MDP Global Championship for a Python/MATLAB sensor-fusion pipeline that cut measurement discrepancies by 20%; delivered 5 digitization projects with SQL and Power BI.

Publications (arXiv preprints, under review)

Event-Driven Safe Control of a Black-Box Balancing Wheel with Friction-Gated Actuation — 2nd author, 2026
YOR: Your Own Mobile Manipulator (navigation stack) — Co-author, 2026

Projects

Go2 RL Locomotion | *Isaac Lab, PPO/RSL-RL, Unitree Go2*
- Isaac Lab PPO locomotion: torque PD control, reward shaping, friction randomization, terrain curriculum (sim-to-real).

CPR Training Machine (team project) | *Raspberry Pi 5, BLDC, FOC, Python/C*
- Co-built a portable AHA-compliant CPR device: FOC torque feedback holds compression depth within 5% of target.

Awards

Best Agentic Use — Innate Robotics Le Robot Hackathon 2025: OpenAI-powered robotic poker player with gesture recognition